

Annexure A

Concept Mapping

Question 1 (5 Marks)

Construct a multi-level concept map integrating:

Operating System Types → OS Structures → System Calls → OS Functions.

Your map must show cross-links (e.g., how OS structure influences system call implementation).

Question 2 (5 Marks)

Design a comprehensive concept map representing:

Process Management, Memory Management, File System, and I/O System.

Include interdependencies and indicate where system calls interact with each component.

Question 3 (5 Marks)

Create a comparative concept map for:

Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.

Highlight differences, similarities, advantages, and limitations using linking phrases.

Question 4 (5 Marks)

Develop a dynamic concept map showing the execution flow:

User Program → API → System Call Interface → Kernel → Hardware → Response back to User.

Include feedback loops and control flow paths.

Question 5 (5 Marks)

Construct a hierarchical + functional concept map for system calls:

Process, File, Device, and Information system calls.

Also show how they relate to OS services like scheduling, memory allocation, and device handling.

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks
Q1						
Q2						
Q3						
Q4						
Q5						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____

→ System Calls → OS functions

1. Operating System Types

Batch OS Jobs are collected in batches and executed	Time-Sharing OS Many users share CPU using time slices.	Distributed OS Manages multiple systems connected over a network	Real-Time OS Responds to events within strict time limits
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OS STRUCTURES

Monolithic All services run in one large kernel	layered OS divided into layers: each layer uses lower ones.	Microkernel Core kernel is minimal: services run in user space.	Modular Kernel has core * loadable modules	Hybrid Combination of monolithic and micro-kernel
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System Calls

Process Control exec(), fork(), exit(), wait()	file Management open(), read(), write(), close()	Device Management ioctl(), read(), write(), close()	Memory Management brk(), mmap(), munmap()	Information getpid(), time(), uname()
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OS functions

Process creates, schedules & terminates processes	Memory Allocates and manages the memory	file System Manages files & directories on storage	User Interface Provides interface b/w user & OS
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Process Management, Memory Management, File System and I/O System

USER APPLICATIONS

System Call Interface

Provides an interface (API) for user programs to request service

Process Management

- Create & delete process
- Synchronization
- IPC
- Deadlock handling

Memory Management

- Tracks memory usage
- Virtual memory
- Swapping
- Protection

File System Management

- File & directory management
- Access control & permissions
- File sharing

I/O SYSTEM Management

- Device allocation
- Buffering & caching
- Device driver management

Interdependencies

- Processes need memory to execute
- Processes read/write files.
- All components use system calls.

Key Takeaway:

All components are interdependent and interact through system calls to provide efficient & reliable.

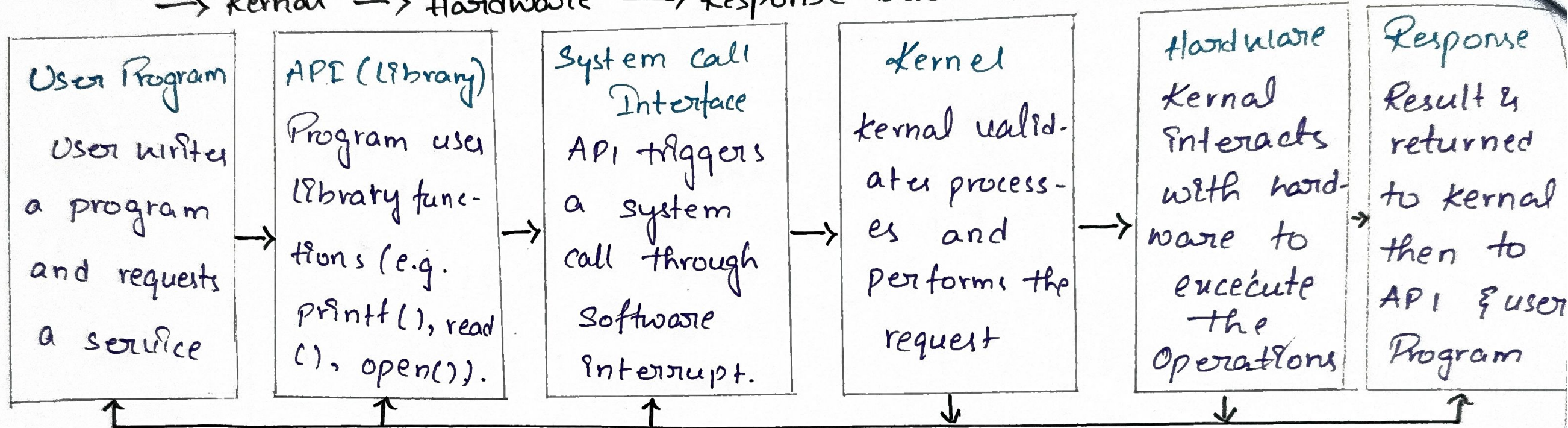
3. Comparative Concept Map of Operating System Types

	Batch OS	Time-sharing OS	Distributed OS	Real-time OS
Definition	Jobs are collected & executed in batches	Multiple users share CPU time interactively	Manage multiple computers as a single system network	Responds to input within strict time deadlines
Key features	- No interaction - User queues	- Interactive - Short response time - Multitasking	- Resource sharing - Scalability - Transparency	- Deterministic - High reliability - Time bound
Advantages	- Simple - Efficient for large jobs - High CPU	- Good response time - Fair CPU usage - User friendly	- High availability - Load sharing - fault tolerance	- Fast response - Reliable - Predictable Performance
Limitations	- No user interaction - long turn around time	- Security issues - Overload of content	- complex communication - security challenges	- Expensive - limited flexibility
Best used for	Payroll, billing, scientific computations	General purpose systems, online services	cloud computing, database systems, enterprise networks	Embedded systems, robotics, aerospace, IOT

Similarities: All manage resources, provide services and support user programs.

Differences: They differ in interaction & response needs.

4. Execution Flow: User Program $\rightarrow$ API $\rightarrow$ System Call Interface $\rightarrow$ Kernel $\rightarrow$ Hardware $\rightarrow$ Response back to User



Return Path: Control & data back in reverse

Feedback Loops:

- Errors or exception return to kernel
- Signals / Interrupts can re-invoke kernel

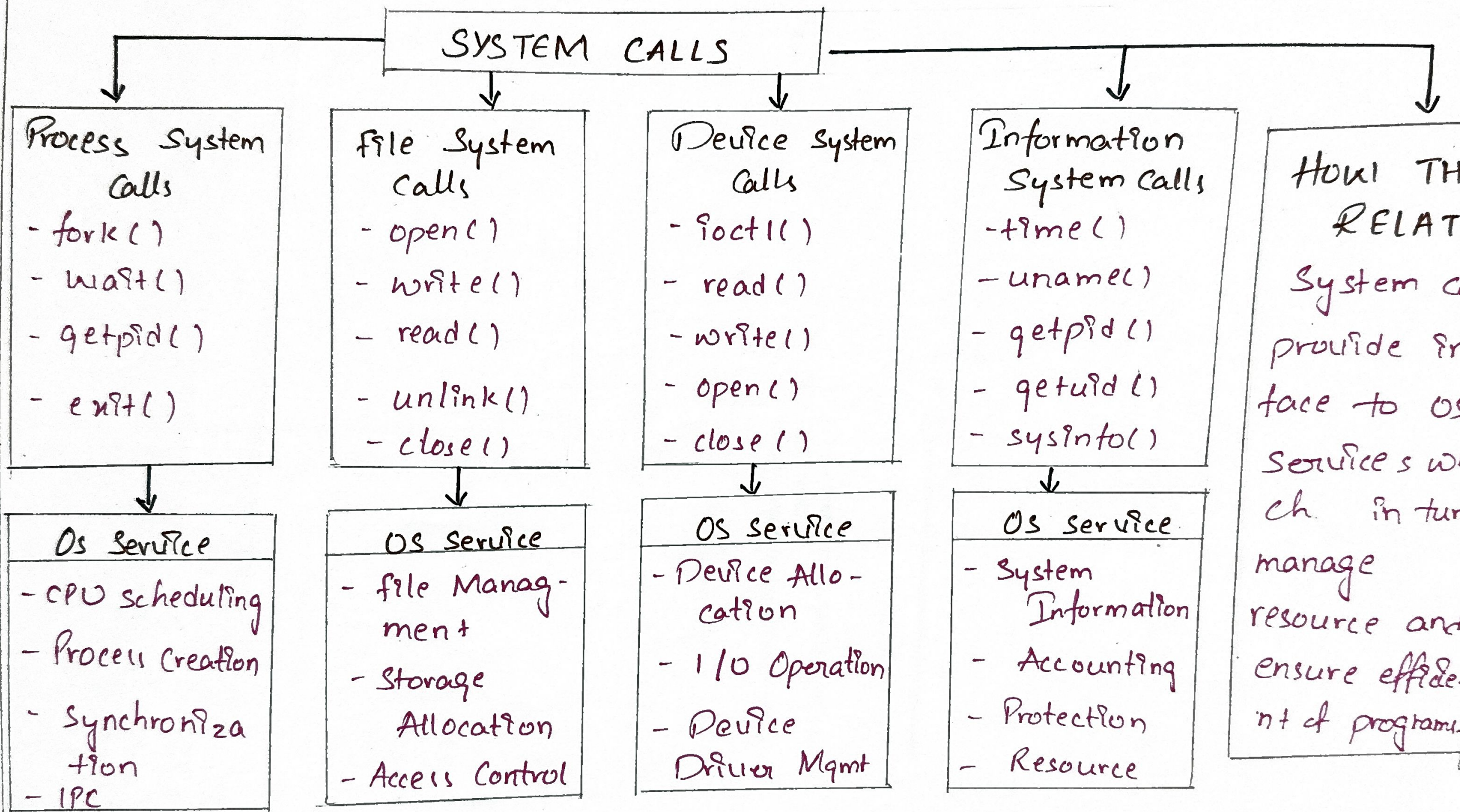
Control Flow Paths

- $\rightarrow$ Normal Flow
- $\dashrightarrow$ Return Flow
- $\dashrightarrow$ Interrupt

KEY TAKEAWAY

System calls acts as a bridge between user programs and hardware through the kernel.

5. System Calls and Related OS Service



Key Takeaway:

System calls are categorised based on their purpose and each group is mapped to OS.